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Mineralization of calcium phosphate in reverse microemulsion

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Abstract

In the present study, reverse microemulsion was prepared to regulate the mineralization of calcium phosphate. Calcium chloride and sodium hydrogen phosphate aqueous solution were, respectively, dropped into the mixture of Span 80, Tween 80, and *n*-heptane to get two kinds of emulsions. *n*-Buranol was used to adjust the emulsions to transparent state. Calcium phosphate was prepared by adding phosphate microemulsion to Calcium microemulsion. Heating method was used to induce the phase separation. After being washed with acetone and ethanol, the mineralized deposition were studied with scanning electron microscopy (SEM), X-ray diffraction (XRD) and transmission electron microscopy (TEM). The results indicated that the minerals in the deposition were mainly monetite (DCPA) and the nanosized needle-like DCPA exhibited the preferential orientation along 200 direction.
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1. Introduction

It has been known that organized assemblies of amphiphilic molecules can be used to provide nanoscale environments for inorganic materials synthesis by mimicking the aspects of biomineralization. Nowadays, these effects have been utilized to prepare useful inorganic materials with desired morphologies in many systems [1–4]. Some amphiphilic molecules can self-assemble to microemulsions under certain conditions, which can provide the needed nanoscale environments for the formation of inorganic materials. Walsh and his colleagues have successfully used bicontinuous reverse microemulsions prepared from the mixtures of didodecyldimethyl ammonium bromide, metastable calcium phosphate aqueous solution and long chain alkane as preorganized systems to fabricate macroporous calcium phosphates [5]. This kind of porous materials has the potential to be used as biomedical implant materials. In the present work, a reverse microemulsion system was used to fabricate the mineralization of calcium phosphate.

2. Materials and methods

In this study, Ca–P minerals were produced by precipitation from microemulsion which prepared with the following procedure: 100 ml *n*-heptane was added into the mixture of Span 80 and Tween 80 (the ratio is 6.7), followed by dropping 10 ml 1 M $CaCl_2$ into the emulsion. About 14 ml *n*-buranol was used to adjust the emulsion to transparent state. During the process, stirring was always needed. Similarly, phosphate microemulsion with

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the content of 6 ml Na_2HPO_4 was also prepared. Calcium phosphate was synthesized by dropping phosphate microemulsion to calcium microemulsion with the reaction time of 10 hours. Heating method was used to induce phase separation. Collected the deposition, then acetone and ethanol was used to wash out the organic component. The sample was subsequently air-dried and studied by SEM, XRD and TEM.

3. Results and discussion

Fig. 1(A) and (B) show SEM photographs of the synthesized calcium phosphate. Besides sheet structure of calcium phosphate, needle-like calcium phosphate, with the length to 14 μm, can be found extensively. The diameter of the needle-like apatite is about 0.18 to 2.1 μm. This displayed that the crystal growth of apatite had been affected by the microemulsion. It is well known that microemulsions can provide nanoscaled reaction environments for nanoscale synthesis or microstructural fabrication. In the present water in oil system, microemulsion trends to have the potential to control the crystal growth along special aspect. From Fig. 1(B), we can find the needle-like apatite was dispersed from aggregated bundles. This is connected with the washing process which was likely to break the intact structure of the calcium phosphate. The lamellar structured apatite marked with dashed circle in Fig. 1(B) was very similar to brushite (DCPD) platelets describe by van Kemenade and de Bruyn [6].

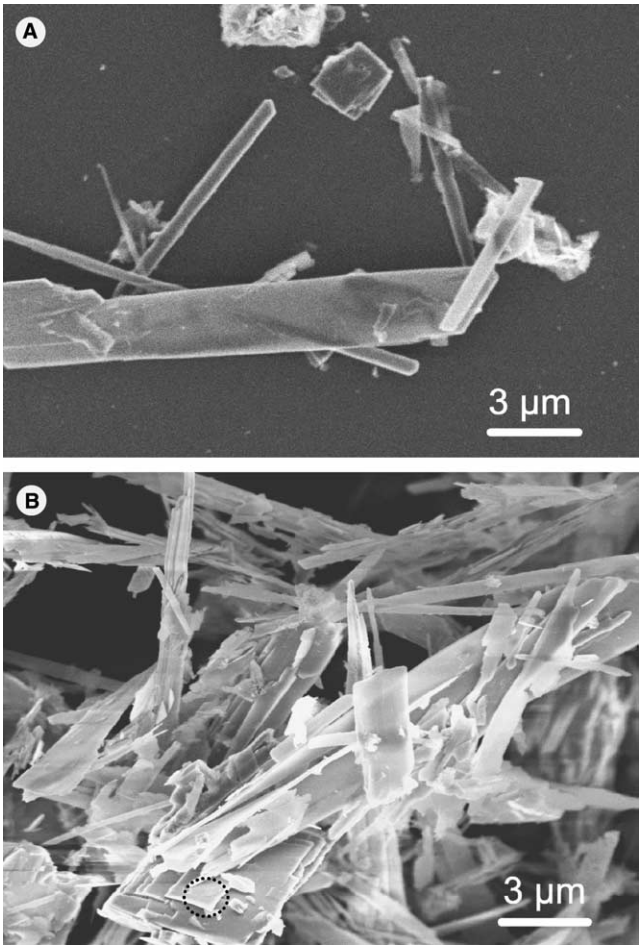

Fig. 1. SEM image of the calcium phosphate from reverse microemulsion.

XRD pattern (Fig. 2) showed that the crystal phase in the synthesized sample was DCPD and DCPA. From the diffraction intensity we can conclude that DCPD predominate in content in the two components.

Fig. 3(A) and (B) are TEM images of the sample which was dispersed by ultrasonic in ethanol and then transferred onto formvarcoated (4% (W/V)) copper grid. Fig. 3(A) and (B) both indicates the diameter of needle-like DCPA can reach to 6–18 nm. SAED patterns exhibit that the preferential orientation exists along 200 direction of DCPA crystal. In Fig. 3(B) the needle-like calcium phosphate are relatively regular in shape. As we know, controlling the nucleation and growth is an important consideration in synthesizing inorganic materials. The SAED results in this research indicated that Span 80 and Tween 80 based microemulsion plays an important role in regulating the crystal growth of calcium phosphate. In Fig. 3(A), little sheet structure of calcium phosphate can be seen. SAED patterns showed DCPA mainly existed in this region, which giving the proof that the little structure was different from the lamellar structure described in Fig. 1(B).

Microemulsions have been used in a number of studies involving biomaterials, semiconductor, catalytic and magnetic materials [7]. Now, the synthesized nanosized apatite also exhibit the good prospect of the new microemulsion system to be used on synthesis of inorganic

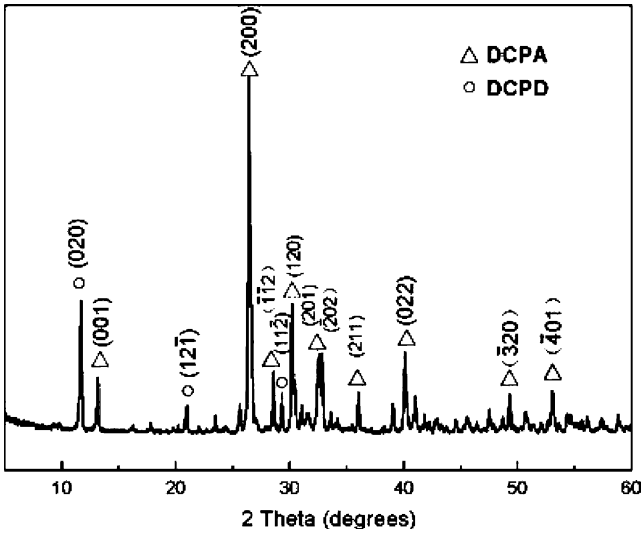

Fig. 2. XRD pattern of the prepared calcium phosphate.

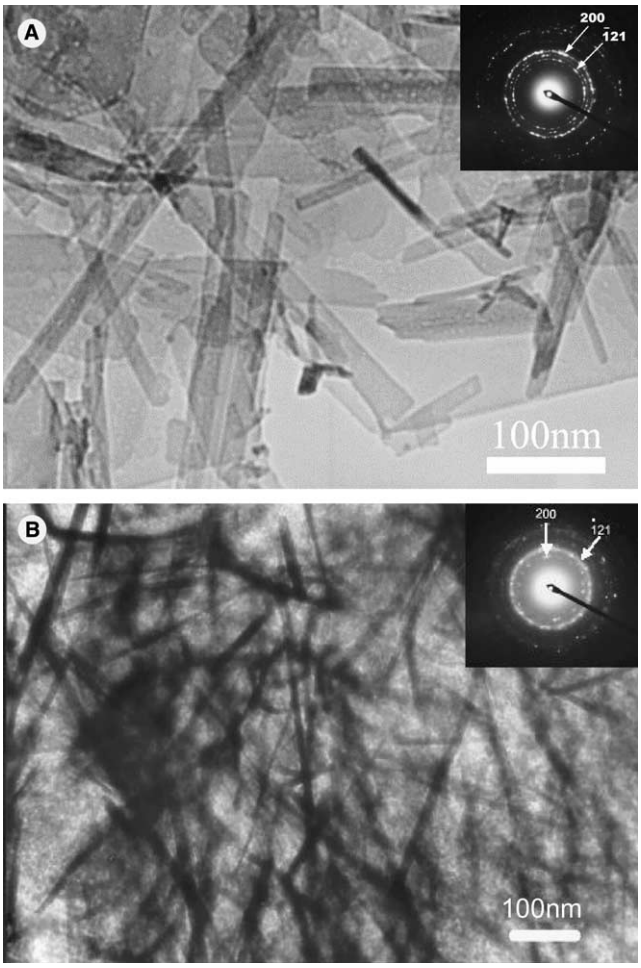

Fig. 3. TEM micrograph of needle-like calcium phosphate. Insets are the SAED patterns from the center of image.

materials, although more detailed work should be done to clarify the mechanism.

4. Conclusions

In this study, we have prepared a new microemulsion system and successfully synthesized nanosized calcium phosphate (DCPA). The diameter of the needles ranges from several to hundreds nanometer and length can reach to 14 μm.. TEM image demonstrates that the needle-like calcium phosphate are regular to some extent in certain region. SAED patterns exhibit the nanosized DCPA has preferential orientation along 200 direction. The prepared microemulsion has the potential to be used on synthesis of inorganic materials.

Acknowledgments

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